Derbyshire Shared Care Pathology Guidelines

Hypomagnesaemia in Adults

Purpose of Guideline
The management of patients with newly diagnosed hypomagnesaemia.

> Magnesium is crucial in the appropriate reabsorption of both potassium and calcium in the kidney. Therefore, in some cases of hypocalcaemia and hypokalaemia it is essential to replace magnesium to enable the correction of both calcium and potassium.

When is hypomagnesaemia considered a medical emergency?

- Magnesium 0.50 – 0.70 mmol/L Not a medical emergency
- Magnesium 0.30 – 0.50 mmol/L Possible medical emergency
- Magnesium <0.30 mmol/L Likely to be medical emergency

These limits are for guidance only; the severity of the patient's symptoms, and other factors, e.g. rapidity of the fall in magnesium, serum calcium and potassium concentrations, renal function, will also determine whether the patient requires emergency admission.

Magnesium results ≤0.40 mmol/L may be telephoned to Derbyshire Health United when the GP practice is closed, dependent on other results.

When is hypomagnesaemia suspected?

Hypomagnesaemia is more common than people think with probably 90% not identified clinically. Symptoms are similar to hypocalcaemia, and for mild hypomagnesaemia are not specific.

Incidence of Hypomagnesaemia

- 2% of the general population
- 7-11% of hospital patients
- Poorly Controlled Diabetes up to 25%
- Critically Ill Patients up to 65%
- Alcoholics up to 80%
- Haematological Malignancies up to 90%

What are the causes of hypomagnesaemia?
Hypomagnesaemia is usually caused by GI and renal loss and redistribution.

Renal causes include:
- Drugs
    - Proton Pump Inhibitors (common cause)
    - Diuretics, especially Loop diuretics
    - Cytotoxic drugs
    - Aminoglycosides
    - Immunosuppressants
    - Theophylline
- Osmotic diuresis (diabetes)
- Inherited Disorders (rare, but include Gitelman's syndrome)

GI causes include:
- Reduced Intake
    - Dietary deficiency (rare)
- Reduced Absorption
    - Diseases causing Malabsorption (e.g. Coeliac Disease)
    - Chronic Diarrhoea
    - Laxative Abuse
    - Fistula
    - Short Bowel Syndrome
    - Inherited Transport Defect (rare)

What happens next?
1. Review the medication - stop PPI's and review diuretics

2. The majority of patients who develop hypomagnesaemia are already under the care of hospital physicians, and referral is usually straight forward (e.g. Oncology)

3. For unknown causes of hypomagnesaemia (<0.50 mmol/L), check U&E and calcium with repeat Mg level.
    - If repeat Mg >0.4 mmol/L and patient is asymptomatic, immediate referral is not required. Consider oral replacement (as below) and referral to appropriate specialty dependent on the patients' history. Endocrine advice is available by choosing 'Advice and Guidance' from NHS e-Referrals.

    - If repeat Mg ≤0.4 mmol/L or with concurrent hypocalcaemia or hypokalaemia, consider referring patient to SDEC

    - Symptomatic patients tend to have more profound hypomagnesaemia (often <0.30 mmol/L) with concomitant hypokalaemia and hypocalcaemia. Consider referral to SDEC in these cases.

Treatment
Treatment of severe hypomagnesaemia is usually intravenous and should always be carried out in hospital. Intramuscular magnesium injections are very painful and are not recommended.

Oral administration
This is used for chronic magnesium loss or moderately severe hypomagnesaemia i.e. where there is a serum magnesium level of approximately 0.4 - 0.7 mmol/L and patients are asymptomatic. Give up to 50 mmol/L day.

Suggested initial treatment is;
- Co-magaldrox 10-20 mL qds (although this product is off-label, this remains the preferred first-line treatment)

During oral replacement monitor response by rechecking magnesium 1-2 weekly initially depending on clinical context. Reversal of hypomagnesaemia with oral replacement may take 6-8 weeks. Long term maintenance replacement may be needed if a reversible cause is not found and removed. In these circumstances, aim to maintain normal serum magnesium using the lowest effective dose of oral replacement.

Diarrhoea tends to limit the amount of magnesium that can be given orally; if diarrhoea develops reduce the dose. The aluminium contained in Mucogel may reduce the chance of diarrhoea.

(Note: The other product occasionally appropriate for use in this context is Magnesium glycerophosphate (1 tablet = 4 mmol Mg, Dose= 1-2 tablets three to four times daily (12-32 mmol/day). Prescription should be reserved for cases of intolerance to Mucogel.)

Patients with Renal Impairment should be treated with caution. Suggest obtain renal advice before commencing treatment if CKD 4 – 5.

Contacts
Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131 (24/7)
Endocrinology Advice	07879 115507 (9am – 5pm, Mon – Fri)
Consultant Phone Triage	Via RDH switchboard, 01332 340131 (10am to 6pm Monday-Friday)
MAU and ACC	01332 788707

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
A&E, EMU and CDU	Via CRH switchboard

Burton

Duty Biochemist	01283 566333 ext 4233
Endocrinology advice	Via endocrine secretaries 01283 593259

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Reviewed by:	Date:	Expiry date:
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Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	June 2016	30th Jun 2018
Dr D Hughes, Dr A Ugur, Dr P Shillo, Dr P Blackwell, Mrs H Seddon	Oct 2021	31st Oct 2023
Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Apr 2024	30th Apr 2026